

1 Configurations Specific to PDV

Overview of mass data archiving

PDV mass data is archived using separate programs and configurations. The entries made within the HYDRA data management are not relevant in this case.

The archiving function of HYDRA-PDV allows for mass data tables to be archived. With this process table data is written in files that are saved in a predefined path.

Mass data tables, also referred to as TNT tables, include a time stamp in their name ID, which is referred to during archiving.

The "tnt_table_repo" table provides information on the archiving status of the single TNT tables, e.g. where the data file is located and what's the file's name or whether the table has been re-imported and thus a new export might be required.

Configuration

Mass data is archived by way of two components: export and import.

The export represents a cyclic process, which is started by the HYDRA Scheduler and unloads "due" tables from the database in data files on the hard disk.

The import (or reload) function constitutes a library that, if required, reloads data on request from the data files back to the database tables where they can be accessed again.

In general, the export function transfers the files 1:1 from the database into the data files.

Program: `hp_mexp.exe / out`

Installation: The program is entered in the HYDRA Scheduler and started cyclically.

Console menu: File – System administration – Scheduler

The corresponding time, when the program is to be started can be defined in the "fix" tab.



Please note: The export program should not be started with little intervals in the Scheduler. As it is an archiving program. Subject to the data volume, daily (every night) or weekly intervals are sufficient.

Moreover, a parameter has to be defined indicating which conditions are required to consider data to be "old enough" to be archived. In this case, the parameter refers to the number of tables that may be online. If this number is exceeded the oldest TNT tables are archived by running an export. The parameter can be found in the "pdv_setup" table and may be adjusted in the basic parameter settings of the console.

As part of the implementation process, it has to be checked whether the archiving path exists with ID "PDVARC" in the "arc_path" column of the "pdv_setup" table (and if applicable, whether the transport path has been defined as well with the "PDVTRANS" ID in the "trans_path" column).

The corresponding directories which are referred to, can be found in the path configuration ("hy_path" table).

A host must not be defined if a relative URL path is defined for archiving, PDVARC, (depending on the file hp_mexp.scr).

Overview: archiving of transaction data

In addition to dynamic TNT tables, movement/transaction data from "fixed PDV tables" is now also archived. In contrast to mass data, this data is archived plus the corresponding configuration of HYDRA data management.

A valid license has to be available on the system to be able to use the archiving function for PDV data.

PLEASE NOTE:



Provided that PDV archiving is enabled and a valid license is not available, corresponding data is deleted from the system, once the specified archiving period has expired (6 months by default).

Data structures

The following PDV data is taken into account for the archiving method described in this section:

- Validity periods of operations relevant to PDV
- Recorded PDV events
- PDV events due to violated limit values
- Modification of default values
- ID tracing (indexing of tags)

By default, this data is archived on the basis of time. Archiving is performed irrespective of whether the corresponding objects (inspection requirements, mass data, operations, etc.) have been archived or not. As an alternative the same mechanisms can also be used to delete the data (instead of archiving).

The listed data is archived separately by default. The following data is concerned:

Data	Source table(s)
Validity periods of operations relevant to PDV	pdv_ag_protokoll
PDV events (recorded events and events caused by violated limit values)	pdv_event_prot
Change of default values	pdv_spc
Indexing of tags	tnt_tagid_repo

Standard configuration

Archiving is configured in two stages by default.

- In the first step data is moved to the medium-term data area. Then data is directly available for evaluations/reports using the medium-term data area.
- In the second stage data is moved to the long-term data area. In this case, data is no longer directly available for evaluations/reports. Data needs to be reloaded first to be able to use it for evaluations/reports in HYDRA.

The default configurations for the first and second archiving level are described in the below table.

Product	Object	Description of the action	Default interval
PDV72	PDV_ADEPRO	Validity periods of operations relevant to PDV are moved from the online dataset to the medium-term dataset	6 months Reference field: abmeld_dat
PDV72	A_PDV_ADEPRO	Validity periods of operations relevant to PDV are moved from the medium-term dataset to the long-term dataset	5 years Reference field: abmeld_dat
PDV72	PDV_EVENT	PDV events are shifted from the online dataset to the medium-term dataset	6 months Reference field: capture_ts
PDV72	A_PDV_EVENT	PDV events are moved from the medium-term dataset to the long-term dataset	5 years Reference field: capture_ts
PDV72	TARGET_VALUE	Changed default values are moved from the online dataset to the medium-term dataset	6 months Reference field: capture_ts

Product	Object	Description of the action	Default interval
PDV72	A_TARGET_VALUE	Changed default values are moved from the medium-term dataset to the long-term dataset	5 years Reference field: capture_ts
PDV72	TNT_TAGID	Tag indexing is moved from the online dataset to the medium-term dataset	6 months Reference field: capture_ts_end
PDV72	A_TNT_TAGID	Tag indexing is moved from the medium-term dataset to the long-term dataset	5 years Reference field: capture_ts_end

Management data of the previously described data will be kept 12 years in the `arc_verw_pdv` table by default.